CityMart Three-Tier Architecture - VPC Networking Implementation

VPC Creation

From the VPC console screenshot:

```
VPC Name: citymart-3-tier-architecture-vpc
VPC ID: vpc-0206bba22a724b50b
IPv4 CIDR: 10.0.0.0/16
Region: eu-west-2 (London)
DNS resolution: Enabled
DNS hostnames: Enabled
```

This configuration forms the network boundary for the entire application infrastructure.

Why this configuration was chosen

The 10.0.0.0/16 CIDR block provides a large address space (65,536 IPs) to support segmentation of the architecture into multiple tiers.

Our architecture definition

- Public web tier
- Private frontend web tier
- Private application tier
- Private database tier

1. Subnet Architecture

From the screenshots, 8 subnets were created across two Availability Zones.

Availability Zones:

```
eu-west-2a
eu-west-2b
```

Total subnets:

```
2 public subnets
6 private subnets
```

Subnet Layout

Tier	Subnet	CIDR	AZ	Purpose
Public	web-public-subnet-1a	10.0.0.0/20	eu-west-2a	Public ALB + NAT
Public	web-public-subnet-1b	10.0.16.0/20	eu-west-2b	Public ALB + NAT
Web	web-private-subnet-2a	10.0.48.0/20	eu-west-2a	Frontend EC2
Web	web-private-subnet-2b	10.0.64.0/20	eu-west-2b	Frontend EC2
App	app-private-subnet-3a	10.0.96.0/20	eu-west-2a	Backend EC2
App	app-private-subnet-3b	10.0.112.0/20	eu-west-2b	Backend EC2
DB	db-private-subnet-4a	10.0.144.0/20	eu-west-2a	RDS
DB	db-private-subnet-4b	10.0.160.0/20	eu-west-2b	RDS

These subnets create clear network isolation between architecture tiers.

2. Internet Gateway Configuration

From the screenshot:

```
Internet Gateway: igw-004e50225c0fdaf27
Name: 3-tier-architecture-sam-igw
State: Attached
```

Step performed

Internet Gateway created.

IGW attached to VPC.

Why this step was required

An Internet Gateway provides connectivity between the VPC and the public internet.

Without this:

the public load balancer

NAT gateways

would not have internet access.

3. NAT Gateway Configuration

From the network connections panel we see:

- `nat-gateway-1`
- `nat-gateway-2`

These NAT gateways were placed in each public subnet.

Implementation steps

Allocate Elastic IP.

Create NAT Gateway in public subnet.

Associate NAT gateway with route tables.

Why NAT Gateways were required

Our architecture uses private subnets for all EC2 servers.

Private instances cannot reach the internet directly.

However, they still need outbound access for:

- package installation
- OS updates
- S3 access
- pulling application code
- AWS Systems Manager

Therefore, traffic flows:

```
Private EC2
↓
Private route table
↓
NAT Gateway
↓
Internet Gateway
↓
Internet
```

Our project documentation confirms two NAT gateways for high availability.

4. Route Table Configuration

From the screenshots we see 8 route tables.

Examples:

```
Public RT
Web-A RT
Web-B RT
App-A RT
App-B RT
Db-A RT
Db-B RT
```

Each subnet is associated with its own route table.

<u>Public Route Table</u>

From the screenshot:

```
Destination: 0.0.0.0/0
Target: igw-004e50225c0fdaf27
```

Meaning:

Public subnets send internet traffic directly to the Internet Gateway.

Used by:

```
Public ALB
NAT Gateways
```

<u>Private Route Tables</u>

Example (DB route table):

```
Destination: 0.0.0.0/0
Target: nat-gateway
```

Meaning:

Private subnets cannot reach the internet directly.

Instead, they send traffic to the NAT Gateway.

5. Resource Map Network Architecture

The VPC resource map confirms the following structure.

```
VPC
 ├── Public Subnets
 │      ├── Public ALB
 │      └── NAT Gateway
 │
 ├── Private Web Subnets
 │      └── Frontend EC2 (Nginx)
 │
 ├── Private App Subnets
 │      └── Backend EC2 (Apache + PHP)
 │
 └── Private DB Subnets
        └── RDS MySQL
```

This segmentation ensures network isolation between tiers.

6. Why Web Servers Were Placed in Private Subnets

Web servers were deployed in private subnets to:

- reduce attack surface
- prevent direct internet access
- enforce traffic control through the ALB

Traffic flow therefore becomes:

```
Internet
↓
Public ALB
↓
Frontend Nginx servers
↓
Internal ALB
↓
Backend PHP servers
↓
RDS database
```

7. Integration with the Application Code

Our networking architecture supports the CityMart application stack.

The CityMart application code is stored in the S3 bucket `3-tier-architecture-sam`, organised into two folders:

```
citymart-frontend/
citymart-backend/
```

This separation enables independent deployment and scaling of the frontend and backend tiers.

The frontend (`citymart-frontend`) contains the static website served by Nginx on frontend EC2 instances, including pages such as `index.html`, `products.html`, `product.html`, and `checkout.html`, along with CSS, JavaScript, and image assets. JavaScript files send API requests to the backend.

The backend (`citymart-backend`) contains the PHP API and business logic running on Apache + PHP on backend EC2 instances. Key API endpoints include:

- `GET /api/health`
- `GET /api/products`
- `GET /api/product?id=1`
- `POST /api/order`

These APIs interact with the Amazon RDS MySQL database (`citymart`), which contains the tables:

- `Products`
- `Customers`
- `Orders`
- `order_items`

During instance startup, EC2 servers use the IAM role *`3-tier-architecture-sam`* to retrieve their respective application code from S3. Frontend servers download `citymart-frontend` and serve it via Nginx, while backend servers download `citymart-backend` and run the API services.

The complete request flow through the architecture is:

```
User → CloudFront → Route53 → Public ALB
→ Frontend EC2 (Nginx)
→ Internal ALB
→ Backend EC2 (Apache + PHP)
→ RDS MySQL (citymart)
```

This architecture implements a scalable three-tier cloud design with separated presentation, application, and database layers, enabling high availability, secure networking, and Auto Scaling support.

8. Domain and Traffic Entry Layer

The external access layer is defined as:

```
User
↓
Route53
↓
citymart.samuelonyejekwe.com
↓
Public ALB (HTTPS)
↓
Frontend EC2
```

From our architecture file:

```
CloudFront Distribution:
d34nv3nql5vjj2.cloudfront.net
```

Route53 resolves our domain to the Public ALB.

9. High Availability Design

Our architecture implements high availability through:

- `2 Availability Zones`
- `Load balancers across AZs`
- `NAT gateways per AZ`
- `Subnets duplicated per tier`

This ensures:

- resilience
- automatic failover
- zero single points of failure

VPC Routing and NAT Gateway Design - CityMart Architecture

In this architecture, our route tables are not associated directly with NAT Gateways. Instead, subnets are associated with route tables, and those route tables contain routes that direct traffic to targets such as NAT Gateways or Internet Gateways.

In other words, each private subnet route table is configured with a default route (`0.0.0.0/0`) that points to a NAT Gateway within the same Availability Zone.

Core Networking Relationships

The routing model in this architecture follows standard AWS networking principles:

- Subnets → associated with route tables
- Route tables → define traffic routes
- Routes → point to targets (IGW, NAT GW)

This distinction is critical for designing scalable and fault-tolerant systems.

Why Two NAT Gateways Were Used

Our architecture includes:

```
natgw1 → Public Subnet (AZ-1)
natgw2 → Public Subnet (AZ-2)
```

This was a deliberate design decision aligned with **high availability and fault isolation**.

<u>Key reasons</u>:

1. ***AZ-Level Fault Isolation (Reliability)***

Each NAT Gateway is deployed per Availability Zone, ensuring:

- If AZ-1 fails → AZ-2 remains operational
- Private subnets in AZ-2 still have outbound internet access

Without this design, a single NAT Gateway would become a **single point of failure**.

2. ***Elimination of Cross-AZ Dependency (Performance + Cost)***

If only one NAT Gateway were used:

- Private subnets in AZ-2 would route traffic to NAT GW in AZ-1
- This introduces:
    - cross-AZ latency
    - inter-AZ data transfer cost

By aligning NAT Gateway per AZ:

```
AZ-1 subnets → natgw1
AZ-2 subnets → natgw2
```

we achieve:

- lower latency
- reduced cost
- better network locality

3. ***Horizontal Scalability (Performance Efficiency)***

NAT Gateways scale automatically, but distributing traffic across two gateways:

- improves throughput distribution
- avoids bottlenecks under load
- supports Auto Scaling in both tiers (frontend/backend)

Why 7 Route Tables Were Used

Our architecture uses:

```
1 Public Route Table
2 Web Tier Route Tables (AZ-1, AZ-2)
2 App Tier Route Tables (AZ-1, AZ-2)
2 DB Tier Route Tables (AZ-1, AZ-2) [optional but structured]
```

Total:

```
7 route tables
```

<u>Reason for This Design</u>

1. ***One Default Route Per Route Table Constraint***

Each route table can only have **one default route (`0.0.0.0/0`)**.

To ensure:

```
AZ A → natgw1
AZ B → natgw2
```

We must create **separate route tables per AZ**.

2. ***Clean Tier Isolation (Security + Operability)***

Each tier has its own route tables:

- Web tier
- Application tier
- Database tier

This allows:

- fine-grained routing control
- easier troubleshooting
- better auditability

3. ***Deterministic Traffic Flow***

Our routing becomes predictable:

Subnet	Route Table	Target
Web-A	Web-RT-A	natgw1
Web-B	Web-RT-B	natgw2
App-A	App-RT-A	natgw1
App-B	App-RT-B	natgw2

Final Routing Model

<u>Public Subnets</u>

```
0.0.0.0/0 → Internet Gateway
```

Used by:

- Public ALB
- NAT Gateways

Private Subnets (Web + App + Database)

```
AZ-1 → 0.0.0.0/0 → natgw1
AZ-2 → 0.0.0.0/0 → natgw2
```

Alignment with AWS Well-Architected Pillars

1. ***Operational Excellence***

- Separate route tables per tier and AZ improve:
    - troubleshooting clarity
    - observability
    - controlled deployments
- Clear routing boundaries align with our structured 3-tier architecture

2. ***Security***

- Private subnets prevent direct internet access
- Traffic flows strictly:

```
ALB → Web → Internal ALB → App → DB
```

- Enforces least privilege networking

3. ***Reliability***

- Two NAT Gateways eliminate single points of failure
- AZ-isolated routing ensures:
    - resilience to AZ outages
    - continued outbound connectivity

4. ***Performance Efficiency***

- AZ-local NAT routing reduces latency
- Avoids cross-AZ traffic overhead
- Supports Auto Scaling efficiently across both AZs

5. ***Cost Optimisation***

- Avoids cross-AZ data transfer charges
- Reduces unnecessary NAT traversal costs
- Enables controlled use of NAT (especially excluding DB tier)

6. ***Sustainability***

- Efficient routing reduces unnecessary network hops
- Lower cross-AZ traffic → reduced energy footprint
- Optimised resource usage aligns with sustainable cloud design

Final Network Architecture

```
VPC (10.0.0.0/16)

 ├── Internet Gateway
 │
 ├── Public Subnets
 │      ├── Public ALB
 │      └── NAT Gateway
 │
 ├── Private Web Subnets
 │      └── Frontend EC2 (Nginx)
 │
 ├── Private App Subnets
 │      └── Backend EC2 (Apache + PHP)
 │
 └── Private DB Subnets
        └── RDS MySQL
```

This networking layer supports the CityMart Online product catalog and order request application.

Summary

Our VPC implementation demonstrates several AWS best practices:

- network isolation by tier
- high availability across AZs

- controlled internet access via ALB and NAT
- private application and database layers
- secure routing between architecture tiers

In this CityMart three-tier architecture, routing is designed using 7 route tables and 2 NAT Gateways distributed across Availability Zones. Each private subnet routes outbound traffic through a zone-local NAT Gateway, ensuring high availability, low latency, and cost-efficient network flow. This design enforces strict tier isolation, eliminates cross-AZ dependencies, and aligns fully with AWS Well-Architected principles, delivering a secure, scalable, and production-grade networking foundation.

cba_admin

VPC > Your VPCs > vpc-0206bba22a724b50b

Transit gateway route tables
Transit gateway multicast
Metering policies

▼ **Traffic Mirroring**

Mirror sessions
Mirror targets
Mirror filters

Network Manager
Cloud WAN
VPC IP Address Manager
AWS Firewall Manager

vpc-0206bba22a724b50b / citymart-3-tier-architecture-vpc

Actions

Details Info

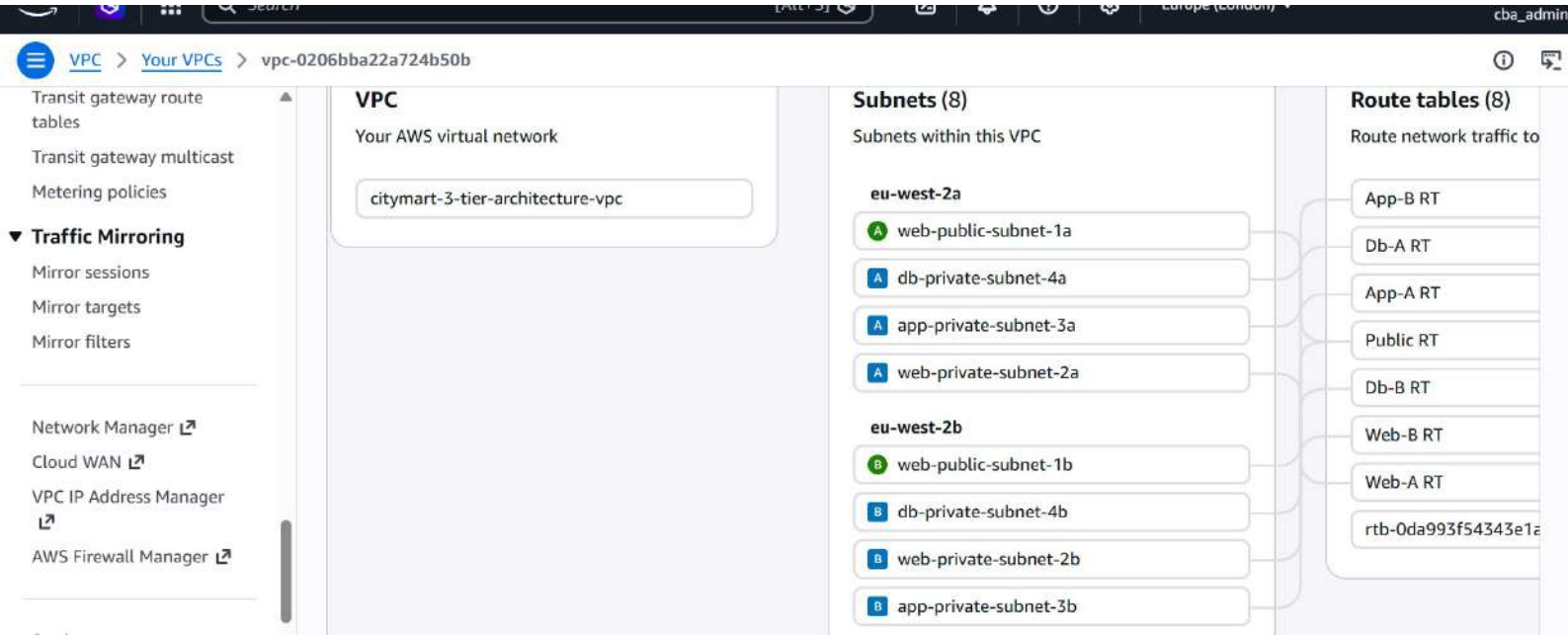
VPC ID vpc-0206bba22a724b50b	**State** Available	**Block Public Access** Off	**DNS hostnames** Disabled
DNS resolution Enabled	**Tenancy** default	**DHCP option set** dopt-01c8cc6efcc8126b4	**Main route table** rtb-0da993f54343e1aa8
Main network ACL acl-0825f6a682570d9e4	**Default VPC** No	**IPv4 CIDR** 10.0.0.0/16	**IPv6 pool** –
IPv6 CIDR (Network border group) –	**Network Address Usage metrics** Disabled	**Route 53 Resolver DNS Firewall rule groups** –	**Owner ID** 009850210027
Encryption control ID –	**Encryption control mode** –		

Resource map | CIDRs | Flow logs | Tags | Integrations

IPv4 CIDRs Info

Edit CIDRs

Address family	CIDR	Status
IPv4	10.0.0.0/16	Associated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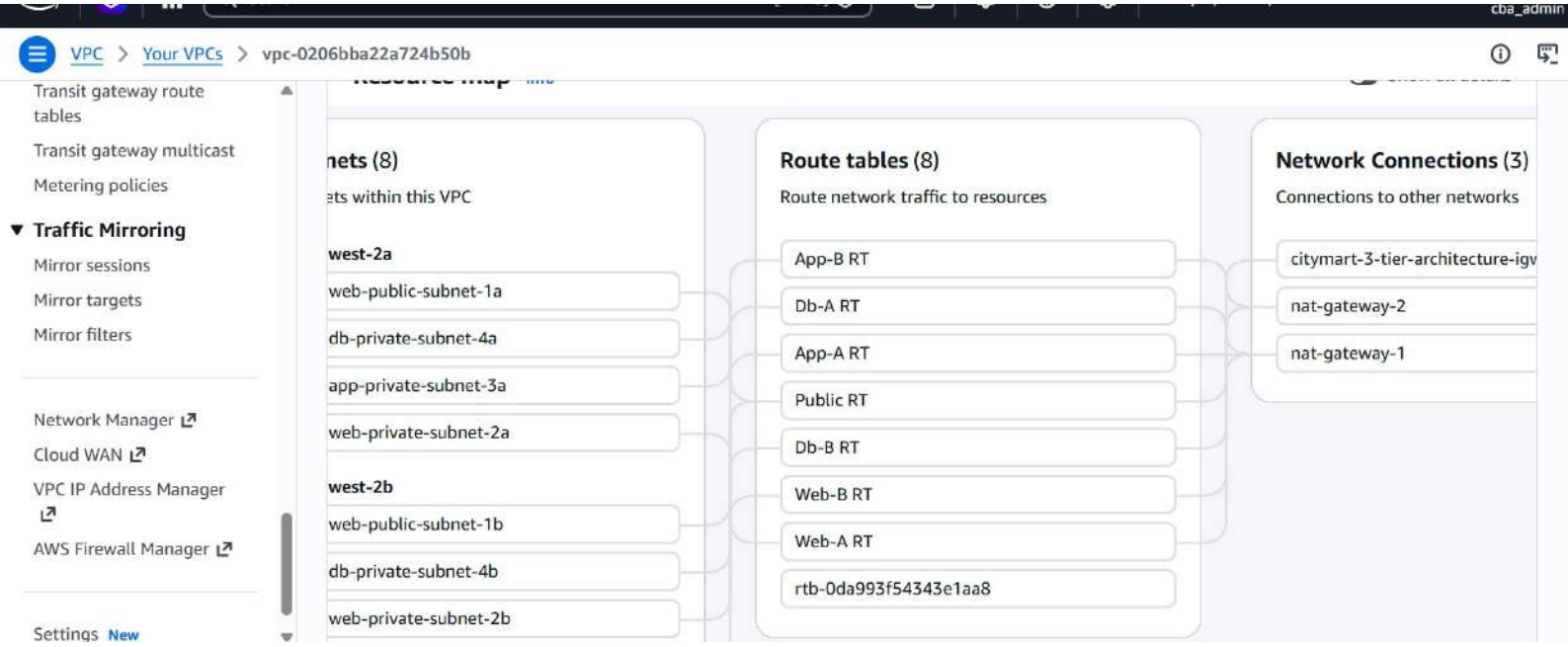
VPC > Your VPCs > vpc-0206bba22a724b50b
Transit gateway route tables
Transit gateway multicast
Metering policies
Traffic Mirroring
Mirror sessions
Mirror targets
Mirror filters
Network Manager
Cloud WAN
VPC IP Address Manager
AWS Firewall Manager
VPC
Your AWS virtual network
citymart-3-tier-architecture-vpc
Subnets (8)
Subnets within this VPC
eu-west-2a
web-public-subnet-1a
db-private-subnet-4a
app-private-subnet-3a
web-private-subnet-2a
eu-west-2b
web-public-subnet-1b
db-private-subnet-4b
web-private-subnet-2b
app-private-subnet-3b
Route tables (8)
Route network traffic to
App-B RT
Db-A RT
App-A RT
Public RT
Db-B RT
Web-B RT
Web-A RT
rtb-0da993f54343e1a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VPC > Your VPCs > vpc-0206bba22a724b50b
Transit gateway route tables
Transit gateway multicast
Metering policies
Traffic Mirroring
Mirror sessions
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Network Manager
Cloud WAN
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AWS Firewall Manager
Settings New
nets (8)
ets within this VPC
west-2a
web-public-subnet-1a
db-private-subnet-4a
app-private-subnet-3a
web-private-subnet-2a
west-2b
web-public-subnet-1b
db-private-subnet-4b
web-private-subnet-2b
Route tables (8)
Route network traffic to resources
App-B RT
Db-A RT
App-A RT
Public RT
Db-B RT
Web-B RT
Web-A RT
rtb-0da993f54343e1aa8
Network Connections (3)
Connections to other networks
citymart-3-tier-architecture-igv
nat-gateway-2
nat-gateway-1

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VPC > Subnets

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Subnets (8) Info

Last updated 11 minutes ago

Actions | Create subnet

Find subnets by attribute or tag

1

Name	Subnet ID	State	VPC
web-private-subnet-2c	subnet-099a048f76a01836f	Available	vpc-0206bba22a724b50b | 3-ti...
web-private-subnet-1b	subnet-0c2983171f07ce599	Available	vpc-0206bba22a724b50b | 3-ti...
web-private-subnet-2a	subnet-0c8a3c4c0ae7b2d7a	Available	vpc-0206bba22a724b50b | 3-ti...
web-public-subnet-1a	subnet-00113ab84ced3cacb	Available	vpc-0206bba22a724b50b | 3-ti...
web-private-subnet-1a	subnet-0056346b219601a91	Available	vpc-0206bba22a724b50b | 3-ti...
web-public-subnet-1b	subnet-07eb876db987e8b02	Available	vpc-0206bba22a724b50b | 3-ti...
web-private-subnet-2b	subnet-05e844ab0f8a66002	Available	vpc-0206bba22a724b50b | 3-ti...

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VPC > Route tables

Transit gateway route tables
Transit gateway multicast
Metering policies
Traffic Mirroring
Mirror sessions
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Mirror filters

Network Manager
Cloud WAN
VPC IP Address Manager
AWS Firewall Manager

Settings New

Route tables (1/8) Info

Last updated 12 minutes ago | Actions | Create route table

Find route tables by attribute or tag

1

Name	Route table ID	Explicit subnet associ...	Edge associations	Main
Public RT	rtb-07cc673de7c73877d	2 subnets	–	No
Db-B RT	rtb-0e0e7ff71fe51b548	subnet-099a048f76a018...	–	No
Web-B RT	rtb-02811d6235299fe5a	subnet-0c2983171f07ce...	–	No
Web-A RT	rtb-0e1f3d735b2473b46	subnet-0056346b219601...	–	No
–	rtb-0da993f54343e1aa8	–	–	Yes

rtb-07cc673de7c73877d / Public RT

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-004e50225c0fd...	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

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VPC > Route tables

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Network Manager
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Settings New

Route tables (1/8) Info

Last updated 13 minutes ago

Actions | Create route table

Find route tables by attribute or tag

1

Name	Route table ID	Explicit subnet associ...	Edge associations	Main
Public RT	rtb-07cc673de7c73877d	2 subnets	–	No
Db-B RT	rtb-0e0e7ff71fe51b548	subnet-099a048f76a018...	–	No
Web-B RT	rtb-02811d6235299fe5a	subnet-0c2983171f07ce...	–	No
Web-A RT	rtb-0e1f3d735b2473b46	subnet-0056346b219601...	–	No
–	rtb-0da993f54343e1aa8	–	–	Yes

rtb-0e0e7ff71fe51b548 / Db-B RT

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-000d6bbbf11c6...	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

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VPC > Route tables

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Route tables (8) Info

Last updated less than a minute ago

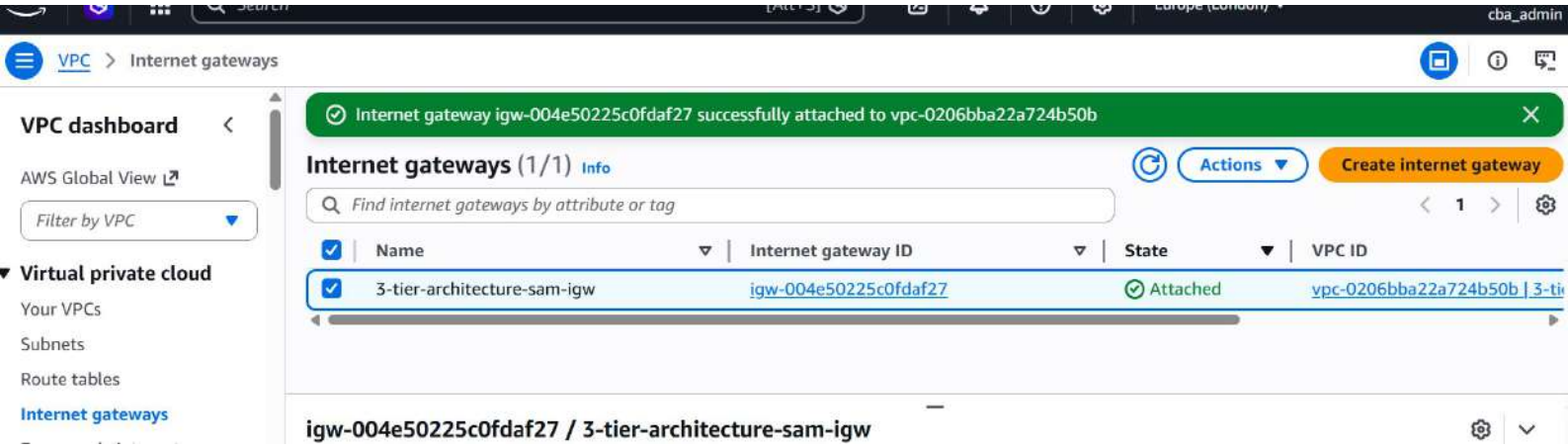
Actions

Create route table

Find route tables by attribute or tag

1

Name	Route table ID	Explicit subnet associ...	Edge associations	Main
Web-B RT	rtb-02811d6235299fe5a	subnet-0c2983171f07ce...	–	No
Web-A RT	rtb-0e1f3d735b2473b46	subnet-0056346b219601...	–	No
Public RT	rtb-07cc673de7c73877d	2 subnets	–	No
Db-B RT	rtb-0e0e7ff71fe51b548	subnet-099a048f76a018...	–	No
Db-A RT	rtb-08f86c30b242a2983	subnet-0bcc97ac1cbd99f...	–	No
App-B RT	rtb-0d8d3f5db11160c37	subnet-05e844ab0f8a66...	–	No
App-A RT	rtb-0245813e6af03f594	subnet-0c8a3c4c0ae7b2...	–	No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VPC > Internet gateways
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Route tables
Internet gateways
Internet gateway igw-004e50225c0fdaf27 successfully attached to vpc-0206bba22a724b50b
Internet gateways (1/1) Info
Actions
Create internet gateway
Find internet gateways by attribute or tag
Name
Internet gateway ID
State
VPC ID
3-tier-architecture-sam-igw
igw-004e50225c0fdaf27
Attached
vpc-0206bba22a724b50b | 3-ti
igw-004e50225c0fdaf27 / 3-tier-architecture-sam-igw

Your VPCs

Subnets

nat-gateway-1	nat-063a3abd490897939	Public	Available	–	Zonal
nat-gateway-2	nat-000d6bbbf11c61d75	Public	Available	–	Zonal